Module5Lab3

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1 Module 5 Lab 3: Textbook Problem 5.5

1.0.1 Age, Income, and Vehicles: What Correlates?

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Repo: https://github.com/raulmckay-blip/cs82a-portfolio

1.1 Overview

This project checks if age, income, and number of vehicles are related.

1.2 Load the data

[1]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

sns.set_theme(style='whitegrid')

cust = pd.read_csv('custdata.tsv', sep='\t')
cols = ['age', 'income', 'num.vehicles']

print(cust.shape)   # 1,000 customers, 11 columns
cust[cols].describe().round(1)
```

```
(1000, 11)
```

[1]:

	age	income	num.vehicles
count	1000.0	1000.0	944.0
mean	51.7	53504.8	1.9
std	18.9	65478.1	1.1
min	0.0	-8700.0	0.0
25%	38.0	14600.0	1.0
50%	50.0	35000.0	2.0
75%	64.0	67000.0	2.0
max	146.7	615000.0	6.0

I started with `describe()` to look for anything off.

1.3 Before cleaning: correlation matrix

[2]:
```
before = cust[cols].corr()
before.round(3)
```

[2]:

	age	income	num.vehicles
age	1.000	0.027	-0.064
income	0.027	1.000	0.139
num.vehicles	-0.064	0.139	1.000

1.4 Finding the invalid values

[3]:
```
print('age = 0:         ', (cust['age'] <= 0).sum())
print('age over 100:    ', (cust['age'] > 100).sum())
print('income below 0: ', (cust['income'] < 0).sum())
print('vehicles missing:', cust['num.vehicles'].isna().sum())
```

```
age = 0:         3
age over 100:    8
income below 0:  1
vehicles missing: 56
```

[4]:
```
# look at the ages over 100
cust.loc[cust['age'] > 100, ['custid', 'age', 'income']]
```

[4]:

	custid	age	income
212	287882	137.700030	60000
266	364991	123.061023	12000
286	397230	136.052160	31200
593	843811	146.680197	40000
625	875771	124.845087	175200
752	1047534	130.401605	200000
926	1303929	126.737284	43400
953	1361012	113.337207	255000

Note: I pulled up the 8 ages over 100. They're all between 113 and 147 and have long decimals (like 137.70003), while every real age in the file is a whole number. That makes them look like errors, not real people.

1.5 Cleaning

What I removed and why:

Impossible values (removed)

- **Age of 0 (3 rows):** No customer is 0 years old, so this is most likely a missing age saved as 0.
- **Age over 100 (8 rows):** Ages from 113 to 147 aren't realistic, and the decimals point to an error.
- **Negative income (1 row):** Income can't be below 0.

Possible values (kept)

- **Income of exactly 0 (78 rows):** That's realistic for someone who's retired, unemployed, or in school.
- **Age of 93 (10 rows):** The data looks like it caps age at 93, so these are real older customers.

Missing values (removed)

- **Missing `num.vehicles` (56 rows):** These aren't wrong, just blank. I dropped them so all three correlations are calculated on the same customers.

[5]:
```
clean = cust[(cust['age'] > 0) & (cust['age'] <= 100) & (cust['income'] >= 0)]
clean = clean.dropna(subset=['num.vehicles'])

removed = len(cust) - len(clean)
print(f'Before: {len(cust)} rows')
print(f'After:  {len(clean)} rows')
print(f'Removed {removed} rows ({removed / len(cust):.1%})')

clean[cols].describe().round(1)
```

```
Before: 1000 rows
After:  932 rows
Removed 68 rows (6.8%)
```

[5]:

	age	income	num.vehicles
count	932.0	932.0	932.0
mean	51.4	55949.3	1.9
std	16.8	66040.1	1.1
min	19.0	0.0	0.0
25%	39.0	17000.0	1.0
50%	50.0	37000.0	2.0
75%	63.0	70000.0	2.0
max	93.0	615000.0	6.0

After cleaning I ran `describe()` again to double-check. Ages now go from 19 to 93 and income starts at 0, so everything left is realistic.

1.6 After cleaning: correlation matrix

[6]:
```
after = clean[cols].corr()
after.round(3)
```

[6]:

	age	income	num.vehicles
age	1.000	-0.003	-0.066
income	-0.003	1.000	0.138
num.vehicles	-0.066	0.138	1.000

1.7 Before vs. after

[11]:
```
pairs = [('age', 'income'), ('age', 'num.vehicles'), ('income', 'num.vehicles')]
compare = pd.DataFrame({
    'pair': [f'{a} vs {b}' for a, b in pairs],
    'r before': [before.loc[a, b] for a, b in pairs],
    'r after':  [after.loc[a, b] for a, b in pairs],
})
compare['change'] = compare['r after'] - compare['r before']
compare.round(3)
```

[11]:

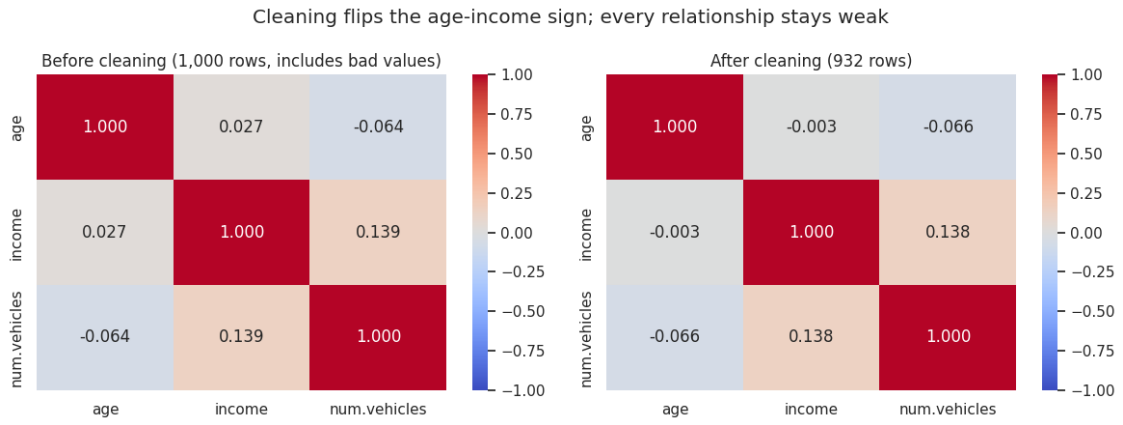
	pair	r before	r after	change
0	age vs income	0.027	-0.003	-0.030
1	age vs num.vehicles	-0.064	-0.066	-0.003
2	income vs num.vehicles	0.139	0.138	-0.001

[8]:
```
fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))

sns.heatmap(before, annot=True, fmt='.3f', cmap='coolwarm', vmin=-1, vmax=1,␣
 ↪ax=axes[0])
axes[0].set_title('Before cleaning (1,000 rows, includes bad values)')

sns.heatmap(after, annot=True, fmt='.3f', cmap='coolwarm', vmin=-1, vmax=1,␣
 ↪ax=axes[1])
axes[1].set_title(f'After cleaning ({len(clean)} rows)')

fig.suptitle('Cleaning flips the age-income sign; every relationship stays␣
 ↪weak')
plt.tight_layout()
plt.show()
```

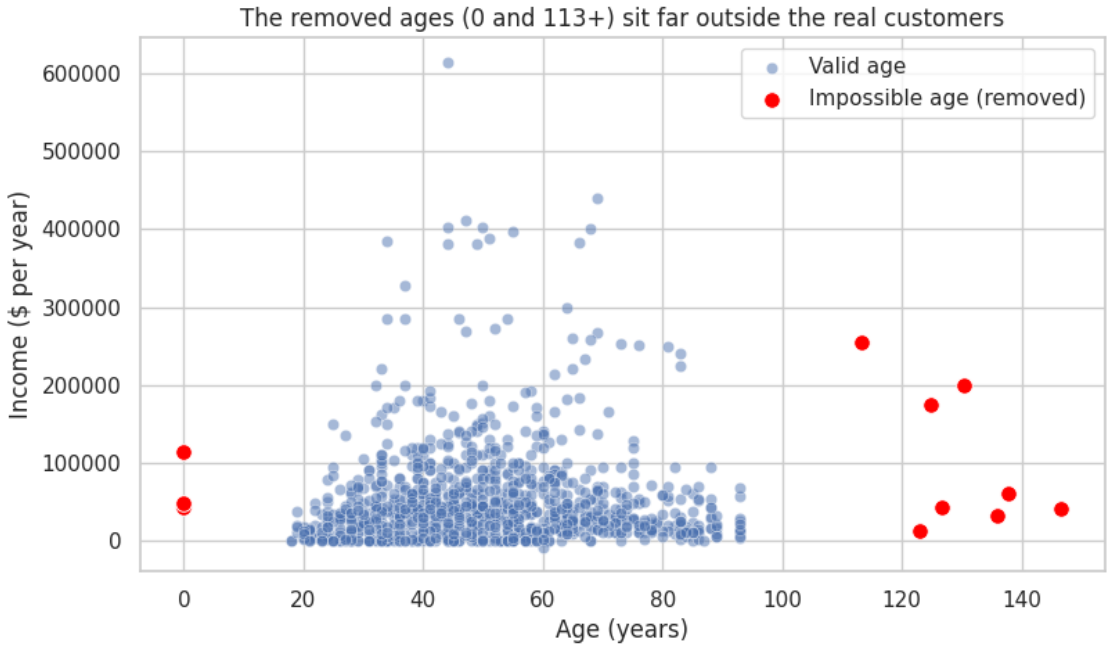


Note: Looking at the two matrices side by side, most numbers barely moved, but age vs. income flipped from +0.027 to -0.003.

1.8 Seeing the bad ages

[9]:
```
bad = (cust['age'] <= 0) | (cust['age'] > 100)

plt.figure(figsize=(9, 5))
sns.scatterplot(data=cust[~bad], x='age', y='income', alpha=0.5, label='Valid␣
↪age')
sns.scatterplot(data=cust[bad], x='age', y='income', color='red', s=70,␣
↪label='Impossible age (removed)')
plt.title('The removed ages (0 and 113+) sit far outside the real customers')
plt.xlabel('Age (years)')
plt.ylabel('Income ($ per year)')
plt.legend()
plt.show()
```



1.9 Checking for a curve ball

Since r only picks up straight-line relationships and since age vs. income came out right around 0, I split age into four groups and checked the average income in each to see if a curve was hiding in there.

[10]:
```
age_groups = pd.cut(clean['age'], bins=[17, 30, 45, 65, 93],
                    labels=['18-30', '31-45', '46-65', '66-93'])
clean.groupby(age_groups, observed=True)['income'].agg(['count', 'mean',␣
↪'median']).round(0)
```

[10]:

age	count	mean	median
18-30	105	29236.0	22000.0
31-45	261	63263.0	45000.0
46-65	376	60952.0	42100.0
66-93	190	50765.0	27750.0

There is one. Income is lowest for 18 to 30, highest from 31 to 65, and drops again after 65. Because it goes up and then back down, a straight line through it is nearly flat, which is why r is close to 0.

1.10 Interpretation (cleaned data only)

Is there a correlation?

Not really a correlation. After cleaning, income and number of vehicles have the strongest relationship, and it's still weak (r = 0.14): people who earn more tend to own slightly more vehicles, with plenty of exceptions. Age has almost no straight-line relationship with income (r 0.00) or vehicles (r = -0.07), but the age-group table shows income does change with age, which r can't pick up.

1.11 AI Use

I used AI to help with syntax and refine wording on this lab.